The Taylor-Doppler Identity:
Frequency Transport on a Phase Manifold

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Abstract

The classical Doppler formula is the first-order term (n=1) of a Taylor expansion of the phase field along an observer's trajectory on a bounded phase manifold. By carrying the expansion to higher orders, we recover the quadratic shell (n=2), whose leading correction structurally corresponds to the second-order term of the relativistic Doppler expansion, and the torsion shell (n=3) as natural consequences of the same mathematical structure. This paper establishes the Taylor-Doppler identity: the Doppler effect, relativistic time dilation, and torsion are not separate physical phenomena but the same expansion evaluated at different shell orders. The derivation does not invoke the postulates of special relativity or predefined coordinate transformations.

1. Introduction

For centuries, physics has treated the Doppler effect and relativistic time dilation as separate phenomena. Each has its own formula, its own interpretation, and its own domain of applicability. This paper demonstrates that they are the same mathematical object: the Taylor expansion of the phase field along an observer's trajectory on a phase manifold.

The classical Doppler formula [8] is the linear shell (n=1). The quadratic shell (n=2) produces a leading correction that structurally corresponds to the second-order term of the relativistic Doppler expansion. The torsion shell (n=3) with its characteristic 1/6 scaling factor corresponds to higher-order phase curvature. Higher-order shells (n≥4) are harmonic overtones.

The Taylor-Doppler identity demonstrates that these phenomena can be represented within a common Taylor expansion framework.

2. The Phase Manifold

Let M be a smooth phase manifold with scalar phase field $\Phi: M \to \mathbb{R}$. The phase is a scalar function of space and time on M.

For an observer at position $x_0 + \Delta x$, the phase is:

$$\Phi(x_0 + \Delta x, t) = \Phi(x_0, t) + (\partial\Phi/\partial x)\Delta x + (1/2)(\partial^2\Phi/\partial x^2)\Delta x^2 + (1/6)(\partial^3\Phi/\partial x^3)\Delta x^3 + \ldots$$

This is the Taylor expansion [7] of the phase field. The stratification of M into harmonic shells is developed in [4] and formalized in [1].

3. Observer Trajectory

The observer moves along a trajectory $x(t)$ on M. The displacement over a small time interval is:

$$\Delta x = v\Delta t + (1/2)a\Delta t^2 + (1/6)j\Delta t^3 + \ldots$$

where v = velocity, a = acceleration, j = jerk. Higher-order terms correspond to snap, crackle, pop, etc.

4. Total Time Derivative on M

The observed frequency is the total time derivative of the phase along the observer's trajectory. Let the observer trajectory be $x(t) = x_0 + vt + (1/2)at^2$. Substituting $(x - x_0) = vt + (1/2)at^2$ into the spatial Taylor expansion yields:

$$\Phi(t) = \Phi_0(t) + (\partial\Phi/\partial x)(vt + (1/2)at^2) + (1/2)(\partial^2\Phi/\partial x^2)(vt + (1/2)at^2)^2 + \ldots$$

Computing the explicit total time derivative $\omega = d\Phi/dt$:

$$\omega = \partial\Phi/\partial t + (\partial\Phi/\partial x) \cdot d/dt(vt + (1/2)at^2) + (1/2)(\partial^2\Phi/\partial x^2) \cdot d/dt(v^2t^2 + vat^3 + (1/4)a^2t^4) + \ldots$$

Executing the differentiation explicitly:

$$\omega = \omega_0 + (\partial\Phi/\partial x)(v + at) + (1/2)(\partial^2\Phi/\partial x^2)(2v^2t + 3vat^2 + a^2t^3) + \ldots$$

Where:

- $\omega_0 = \partial\Phi/\partial t$ = source frequency
- $k_0 = \partial\Phi/\partial x$ = wave number
- $k_1 = (1/2)\partial^2\Phi/\partial x^2$ = phase curvature
- $k_2 = (1/6)\partial^3\Phi/\partial x^3$ = phase torsion

The curvature-frequency relationship $f = c/r$ governing shell transitions is established in [2].

5. The Boundary Projection Operator Ĝ

To connect the continuous phase expansion to discrete shell transitions, we introduce a first-order boundary linearization operator Ĝ.

> **Definition 5.1.** Let $\hat{G}: C^\infty \to P^1$ be a projection operator that maps a continuous transcendental function onto its first-order polynomial shell equivalent at the boundary threshold:

$$\hat{G}[e^{\lambda \Delta t}] \equiv 1 + \lambda \Delta t$$

The operator maps C^∞ functions on a bounded interval to P^1, the space of linear polynomials. It enforces the structural boundary condition where the continuous integration window closes, preventing phase debt accumulation as formally defined in [6].

6. The Exponential Profile

We define a uniform transformation rate as a condition where the fractional rate of change of the boundary state is constant over the evaluation interval:

$$\dot{a}(t) / a(t) = \lambda = \text{constant}$$

Integrating this first-order differential equation:

$$\int (1/a)\, da = \int \lambda\, dt \Longrightarrow \ln|a| = \lambda t + C$$

Setting the initial boundary condition $a(0) = 1$ yields:

$$a(t) = e^{\lambda t}$$

This establishes the exponential form as the unique mathematical consequence of the uniform fractional rate condition. The uniform-rate condition $\dot{a}/a = \lambda$ remains the premise; the exponential is its derived consequence.

7. Shell Mapping

Within the Harmonic Framework, these arise as successive shell orders of a common Taylor expansion. The shell structure and 1/6 scaling properties are developed in [3]. The Law of Admissibility [6] establishes structural completion conditions for recursive systems.

Shell	**Mathematical Term**	**Physical Effect**
$n = 0$	ω_0	Rest frequency
$n = 1$	$k_0 v$	Classical Doppler [8]
$n = 2$	$k_1\, a\Delta t$	Relativistic Doppler / Acceleration [9]
$n = 3$	$k_2\, j\Delta t^2$	Torsion / Higher-order phase curvature
$n \geq 4$	Higher derivatives	Higher-order corrections

8. The Classical Doppler as a Special Case

When motion is uniform and the medium is locally flat, all higher-order derivatives vanish:

$$a = 0,\ j = 0,\ \partial^2\Phi/\partial x^2 = 0,\ \partial^3\Phi/\partial x^3 = 0,\ \ldots$$

The expansion truncates at $n = 1$:

$$\omega = \omega_0 + k_0 v$$

Substituting $k_0 = \omega_0 / v_p$:

$$\omega = \omega_0 (1 + v/v_p)$$

This is the classical Doppler formula [8]. It is valid only when higher-order terms vanish.

9. The Lorentz Factor as the Quadratic Shell

The relativistic Doppler formula [9] is:

$$f\_obs = f_0 \sqrt{((1 + v/c) / (1 - v/c))}$$

Expanding:

$$f\_obs = f_0 (1 + v/c + (1/2)(v/c)^2 + (1/2)(v/c)^3 + \ldots)$$

The $(1/2)(v^2/c^2)$ term is the quadratic shell ($n=2$). The quadratic shell produces the leading quadratic frequency correction that structurally corresponds to the second-order term of the relativistic Doppler expansion.

This quadratic shell corresponds to the bending shell described in [3].

10. The Taylor-Doppler Identity

$$\omega = \omega_0 + \Sigma(n=1 \text{ to } \infty) (1/n!) (\partial^n\Phi/\partial x^n) \, d/dt(\Delta x^n)$$

This expansion produces:

- $n = 1$: Classical Doppler [8]
- $n = 2$: Relativistic Doppler [9]
- $n = 3$: Torsion shell
- $n \geq 4$: Higher-order corrections

The zero-drift validation protocols that ensure structural integrity at each shell are developed in [5, 6].

11. Quadratic Shell Correspondence

The emission-reception cycle consists of two sequential boundary stages: the signal is emitted at the source shell, propagates through the manifold, and is received at the observer’s boundary shell. The sequential composition of these stages is not introduced here as a new assumption. It follows directly from the monadic bind operation established in the Blossom Harmonic Algorithm as Holonomic Monad [5], where admissible tier transitions compose sequentially, each validating and advancing through a distinct boundary before the next stage proceeds.

Each admissible boundary transition inherits the exponential scale profile derived in Section 6. Under the monadic transport formalism of [5], successive boundary transitions compose sequentially rather than superpose, so each contributes one projected transport factor. Under the boundary projection operator Ġ, each factor projects to its first-order polynomial equivalent:

$$f_r = f_0 \cdot \dot{G}[e^{\wedge}(\lambda\Delta t)] \cdot \dot{G}[e^{\wedge}(\lambda\Delta t)]$$

$$= f_0(1 + \lambda\Delta t)^2 = f_0(1 + 2\lambda\Delta t + \lambda^2\Delta t^2)$$

The relativistic Doppler expansion for a kinematic system yields:

$$f\_rel = f_0 \,(1 + \beta + (1/2)\beta^2 + \ldots),\ \beta = v/c$$

The coefficient sequences are not identical, and Δt is retained here as a free transport parameter rather than normalized away. We do not claim algebraic equivalence. Both frameworks generate a natural quadratic correction tier (λ^2 versus $(1/2)\beta^2$) as an inherent structural requirement to account for non-flat field variations. The difference in the linear coefficient (2λ versus β) reflects the dual-boundary interaction of the integrated phase field versus the single coordinate transformation of special relativity.

The parameter $\lambda = v/v_p$ is the normalized velocity of the observer relative to the propagation speed of the medium, while $\beta = v/c$ is the normalized velocity relative to the speed of light in vacuum. They are not the same physical quantity. One describes transport through a medium-defined manifold, the other through vacuum coordinate transformations. Both serve as the expansion parameter for their respective frameworks, producing structurally analogous quadratic corrections in the frequency shift.

This is a structural analogy, not an algebraic equivalence.

12. Numerical Verification

To verify the structural correspondence of the framework against classical kinematic tracking methods, we evaluate an acoustic tracking configuration. Let a stationary acoustic source in a uniform medium emit a pure tone under the following boundary constraints:

Source frequency: $\omega_0 = 1000.0$ Hz
Medium propagation speed: $v_p = 343.0$ m/s
Observer snapshot velocity: $v = 34.3$ m/s ($\lambda = v/v_p = 0.10$)
Observer snapshot acceleration: $a = 9.81$ m/s^2
Parameter evaluation window: $\Delta t = 0.01$ s

12.1 Classical Kinematic Method

The traditional kinematic approach treats acceleration as a first-order correction to the instantaneous velocity over the interval:

$$\omega\_kin = \omega_0(1 + v/v_p) + \omega_0(a\ \Delta t / v_p)$$

$$= 1000.0(1 + 0.10) + 1000.0(9.81 \times 0.01 / 343.0) = 1100.0 + 0.286 = 1100.286 \text{ Hz}$$

12.2 Phase Manifold Framework Method

Under the transport dynamics on M, the frequency is determined by summing the localized derivative shell coefficients over the parameter integration window:

$$\omega\_frame = \omega_0 + k_0\ v + k_1(a\ \Delta t)$$

$$= 1000.0(1 + 0.10) + 1000.0(9.81 \times 0.01 / 343.0) = 1100.0 + 0.286 = 1100.286 \text{ Hz}$$

12.3 Result

$$\omega\_kin = \omega\_frame = 1100.286 \text{ Hz}$$

This confirms that coordinate-free phase field expansion maps perfectly to traditional vector tracking methods at low-dimensional limits. The framework reproduces classical kinematics exactly when higher-order terms vanish. The difference is not in the numbers. It is in the interpretation and the ability to extend to higher-order shells.

12.4 Higher-Order Verification

To verify that the framework extends naturally to higher-order shells, we evaluate a configuration that includes jerk (third-order motion):

Source frequency: $\omega_0 = 1000.0$ Hz
Medium propagation speed: $v_p = 343.0$ m/s
Observer snapshot velocity: $v = 34.3$ m/s ($\lambda = 0.10$)
Observer snapshot acceleration: $a = 9.81$ m/s^2
Observer snapshot jerk: $j = 2.0$ m/s^3
Parameter evaluation window: $\Delta t = 0.10$ s

12.4.1 Classical Kinematic Method

$$\omega\_kin = \omega_0(1 + v/v_p) + \omega_0(a\ \Delta t / v_p) + \omega_0((1/2)\ j\ \Delta t^2 / v_p)$$

$$= 1100.0 + 2.860 + 0.029 = 1102.889 \text{ Hz}$$

12.4.2 Phase Manifold Framework Method

$$\omega\_frame = \omega_0 + k_0\ v + k_1(a\ \Delta t) + k_2(j\ \Delta t^2)$$

$$= 1000.0 + 100.0 + 2.860 + 0.029 = 1102.889 \text{ Hz}$$

12.4.3 Result

$$\omega\_kin = \omega\_frame = 1102.889 \text{ Hz}$$

This demonstrates that the framework extends naturally to higher-order shells. In conventional treatments, these higher-order contributions are typically expressed through successive kinematic derivatives. In the present framework, they arise as successive shell orders of a single Taylor expansion of the phase field.

12.5 Complete Shell-Order Summary

Shell	Name	Kinematic Formula	Framework Formula	Result
$n = 1$	Classical Doppler	$\omega_0(1 + v/v_p)$	$\omega_0 + k_0 v$	1100.000 Hz
$n = 2$	Relativistic Doppler	$+ \omega_0(a\Delta t/v_p)$	$+ k_1(a\Delta t)$	1102.860 Hz
$n = 3$	Torsion	$+ \omega_0((1/2)j\Delta t^2/v_p)$	$+ k_2(j\Delta t^2)$	1102.889 Hz
$n \geq 4$	Higher-order	$+ \omega_0$(higher deriv.)	$+ k_n(\bullet\bullet\bullet)$	generalizes

The pattern established at $n = 1$, 2, and 3 generalizes to all higher orders $n \geq 4$ by the same Taylor expansion, as demonstrated in Section 10. The framework is not limited to the orders shown; it is a complete, scalable method for frequency transport on a phase manifold.

13. Discussion: Kinematic Projections

Classical and relativistic frameworks infer a spatial velocity vector v from an observed frequency alteration Δf. By proving that a graded phase manifold naturally transports frequencies through successive derivative shells, this framework offers an alternative perspective: the frequency shift is the primary operational reality on M, while the kinematic velocity vector is a low-dimensional coordinate projection [10].

This perspective does not invalidate the standard Doppler formula. It contextualizes it as the $n=1$ truncation of a more general expansion. When higher-order terms are present, the velocity interpretation becomes an approximation that omits structural contributions from acceleration and curvature.

14. Conclusion

The classical Doppler formula is the first-order term of a Taylor expansion of the phase field on a phase manifold. The quadratic shell produces the leading correction that structurally corresponds to the second-order term of the relativistic Doppler expansion. The torsion shell is the third-order term. Within the Harmonic Framework, these correspond to successive shell orders of a common Taylor expansion.

The Taylor-Doppler identity demonstrates that these phenomena can be represented within a common Taylor expansion framework. The math is the same. The only difference is how many terms we carry.

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